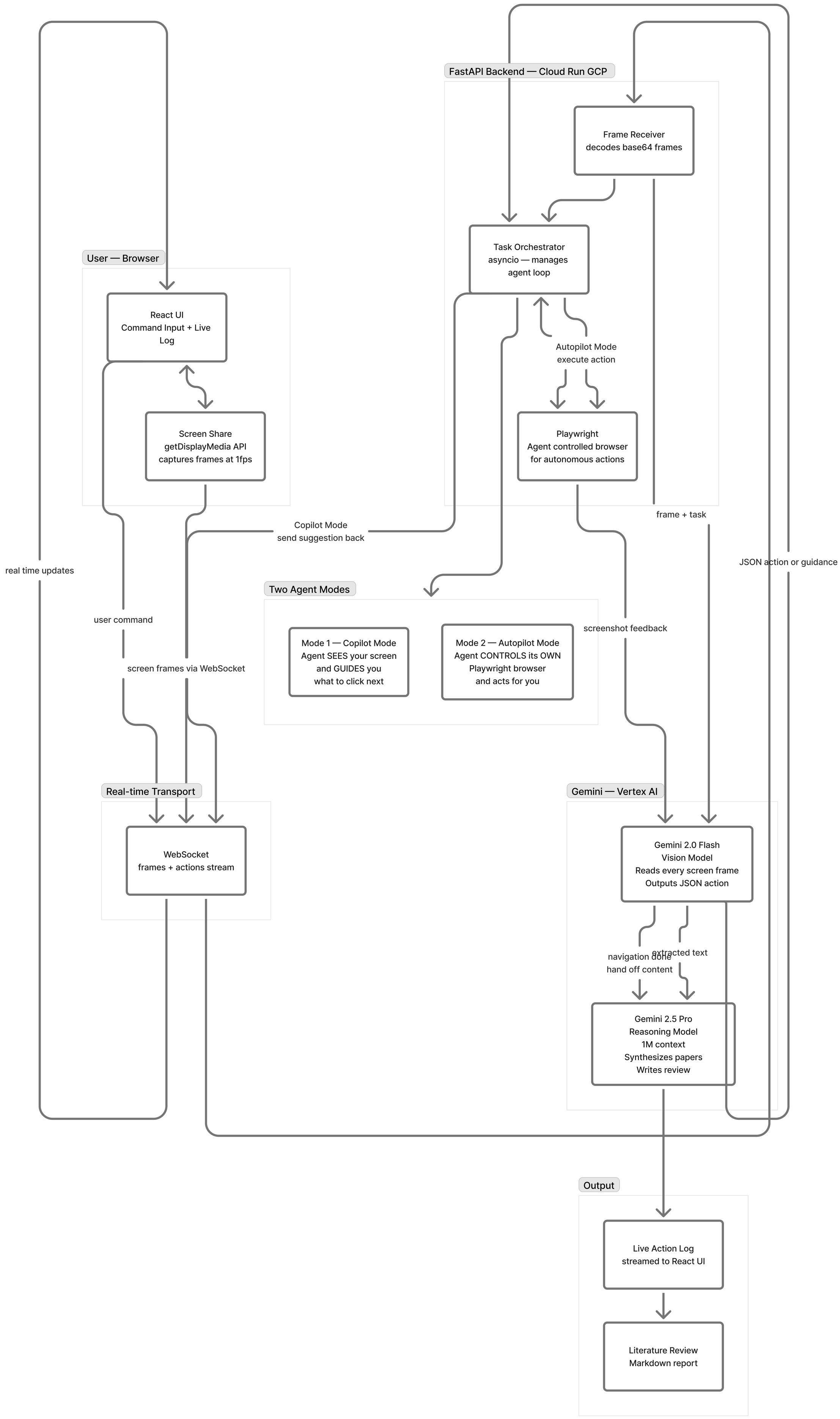
FastAPI Backend — Cloud Run GCP
Frame Receiver
decodes base64 frames
Task Orchestrator
asyncio — manages
agent loop
Autopilot Mode
execute action
Playwright
Agent controlled browser
for autonomous actions
User — Browser
React UI
Command Input + Live
Log
Screen Share
getDisplayMedia API
captures frames at 1fps
frame + task
Copilot Mode
send suggestion back
real time updates
JSON action or guidance
Two Agent Modes
Mode 1 — Copilot Mode
Agent SEES your screen
and GUIDES you
what to click next
Mode 2 — Autopilot Mode
Agent CONTROLS its OWN
Playwright browser
and acts for you
user command
screenshot feedback
screen frames via WebSocket
Real-time Transport
WebSocket
frames + actions stream
Gemini — Vertex AI
Gemini 2.0 Flash
Vision Model
Reads every screen frame
Outputs JSON action
navigation done
hand off content
extracted text
Gemini 2.5 Pro
Reasoning Model
1M context
Synthesizes papers
Writes review
Output
Live Action Log
streamed to React UI
Literature Review
Markdown report